Name [redacted] **ID#** [redacted]

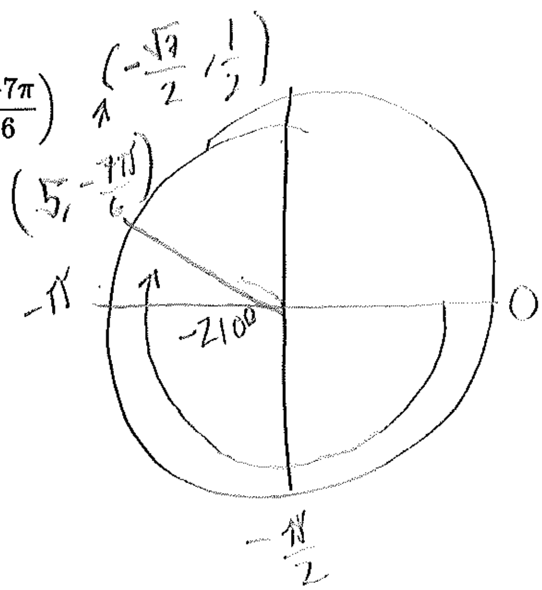
Math 104 Spring 2025 Final Exam

- A non graphing calculator is allowed
- Must show all work to receive credit
- Written or printed notes allowed (notes on digital devices not allowed)
- Attach all scratch paper

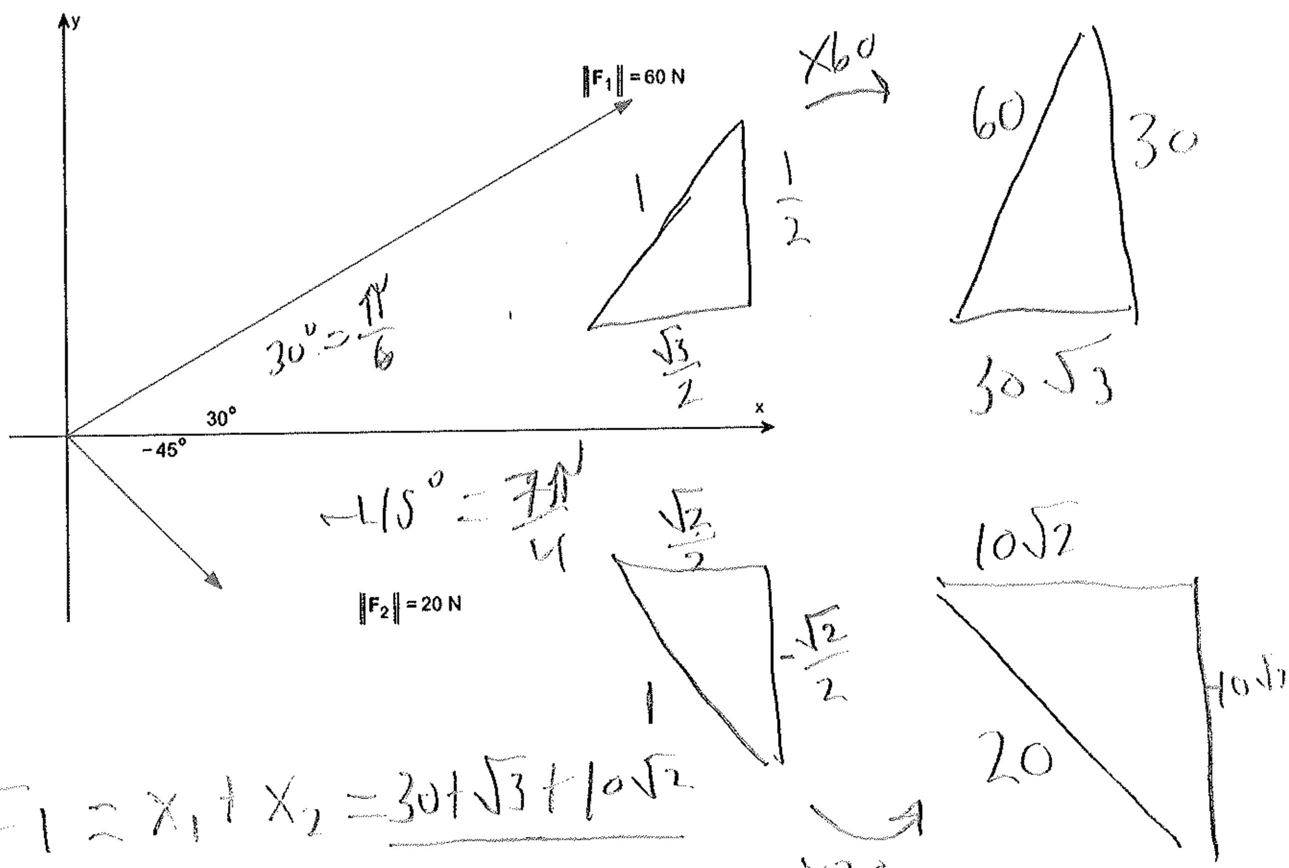
Question		Points
1	Polar, Vectors	/4
2	Average Rate of Change	/4
3	Trig Equations	/6
4	Side Side Angle Triangles	/6
5	Graphing, Domain, Range	/6
6	Applications	/6
Total		/30

SSA or SAS

1a) Plot the point is given in polar coordinates $(r, \theta) = \left(5, \frac{-7\pi}{6}\right)$



1b) Add the two vectors $F_1 + F_2$



$F_1 = x_1 + x_2 = \underline{30 + \sqrt{3} + 10\sqrt{2}}$

$F_2 = y_1 + y_2 = 30 - 10\sqrt{2}$

$F_1 + F_2 = (30 + \sqrt{3} + 10\sqrt{2}) + (30 - 10\sqrt{2})$

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2) Let $g(x) = 4 - x^2$.

a) Find the average rate of change from -1 to 3.

rate of change formula $= \dfrac{f(b) - fa}{b - a}$

4- [3] 1.1= 1

a=-1 $4-(-1)^2=3$ $\quad -5-3=-8 \quad \dfrac{-8}{4} = \boxed{-2}$

$3+1=4$

(3,3)

$4-(3)^2=-5$

4-9

b) Find an equation of the secant line containing $(-1, g(-1))$ and $(3, g(3))$.

$y - fa = -2(x-a)$ | double check!

$y+5=-2(x-3)$

$y-3=-2(x+1)$

$-2x+6-5$

$= -2x-2+3$

$y=-2x+1$

$y=-2x+1$

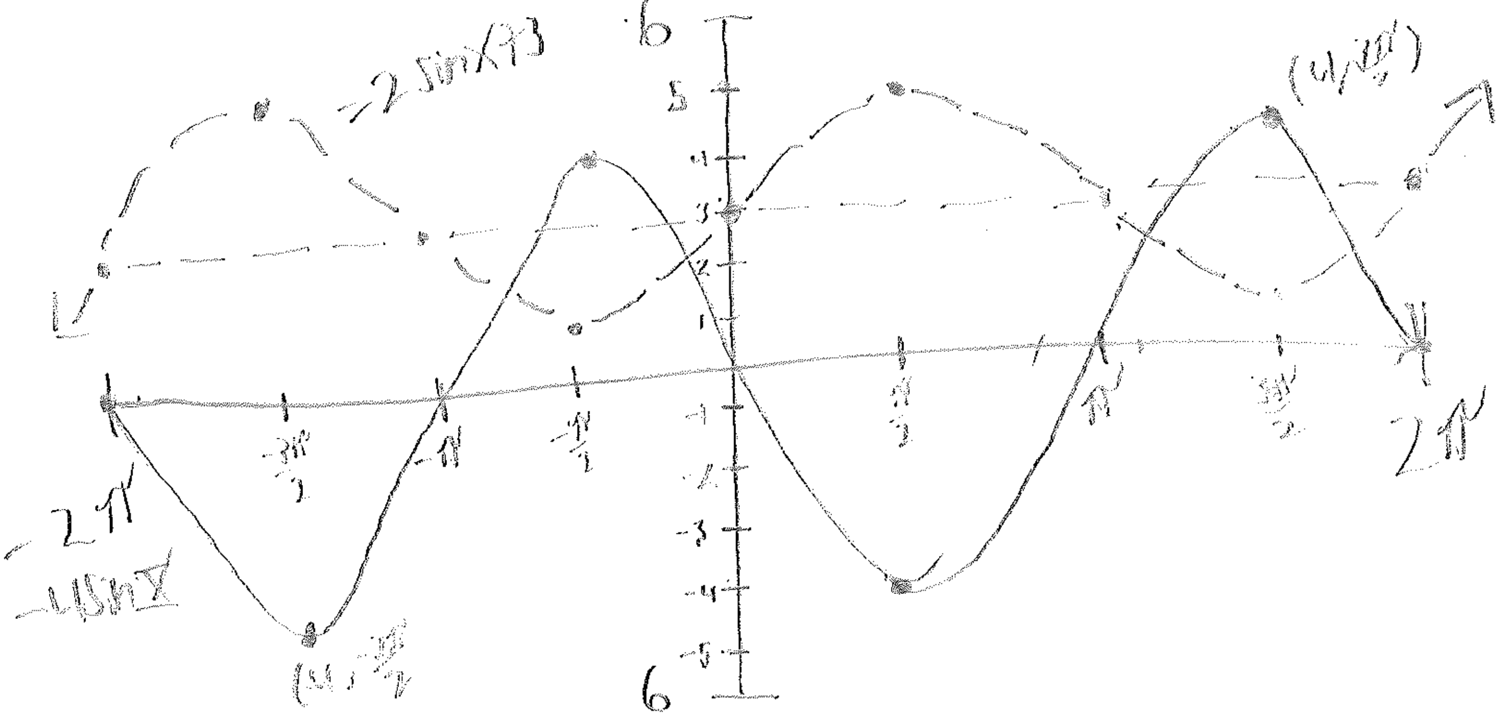
$y=-2x+1$

sin:

-sin: π 2π

3) Let $f(x) = -4\sin(x)$ and let $g(x) = 2\sin(x) + 3$.

Graph $f(x)$ and $g(x)$ on the same axes.



Solve $f(x) = g(x)$ on the interval $[0, 2\pi]$.

$$\frac{-4\sin = 2\sin + 3}{-2\sin \quad -2\sin}$$

$$\frac{-6\sin}{-6} = \frac{3}{-6}$$

$$\sin = \frac{3}{-6} = -\frac{1}{2} =$$

props of $\sin(-\frac{1}{2})$

$$\boxed{\frac{7\pi}{6}, \frac{11\pi}{6}}$$

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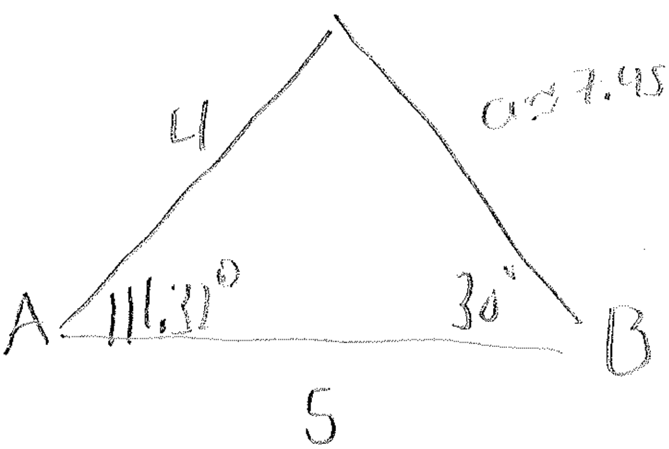
4) Two sides an an angle are given. Determine whether the given information results in one triangle, two triangles, or no triangle at all. Solve any resulting triangle(s) and draw the triangle(s).

$180-30-38.68$

$b = 4, c = 5, B = 30^\circ$

S S A

$C = 38.68$



$$\frac{\sin 30}{4} = \frac{\sin C}{5}$$

$$\sin^{-1}\left(\frac{5\sin 30}{4}\right) = 38.68$$

1st triangle
$C_1 \approx 38.68^\circ$, $A_1 \approx 111.32^\circ$
$a_1 \approx 7.45$

$a = \sqrt{5^2+4^2-2(5)(4)\cos(111.32)}$

$a = 7.45$

Solve 2nd triangle

$180-38.68$

there is a 2nd!

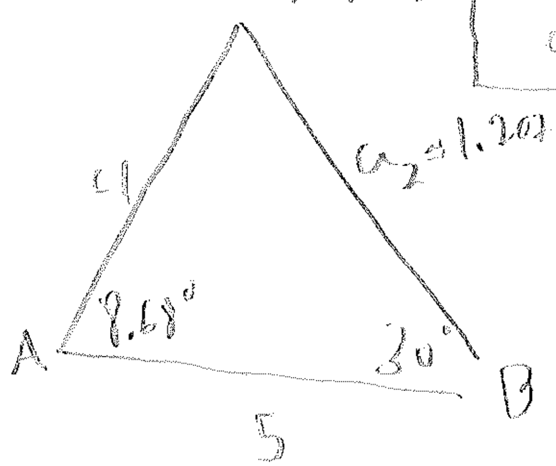
$141.32 + 30 = 171.32$

$180-171.32$

8.68

$C = 141.32^\circ$

2nd triangle
$C_2 \approx 141.32^\circ$, $A_2 \approx 8.68^\circ$
$a_2 \approx 1.207$

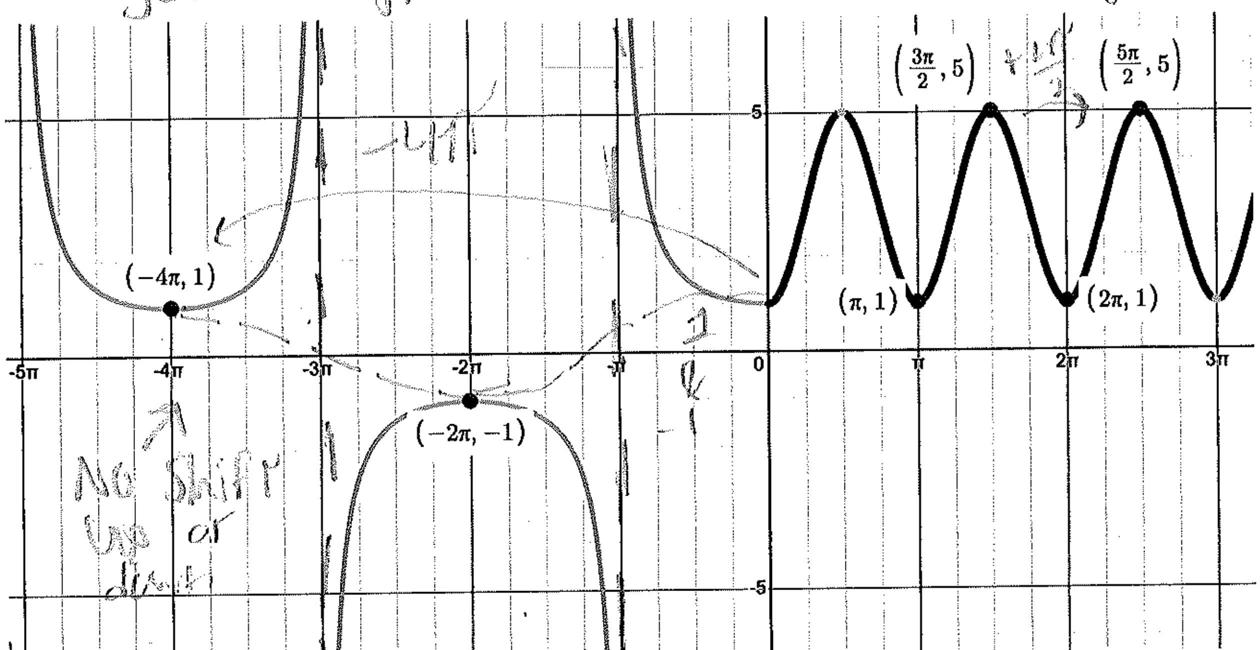


$a = \sqrt{5^2+4^2-2(5)(4)\cos(8.68)}$

$a = 1.207$

5) Given the graph below find the function $f(x)$ and state the domain and range.

Secant graph!　　　　　　　　　　-cosine graph



$\frac{2\pi}{-4\pi} = -\frac{1}{2}$

$\frac{2\pi}{\frac{2\pi}{2}} = \frac{2\pi}{1} \cdot \frac{2}{2\pi} = \frac{4}{2} = 2$

<0　$\underline{1 \sec(-\frac{1}{2}x)}$

range

$[-\infty, -1] \cup [1, \infty]$

Domain

$(-\infty, 0] \cap (K - \frac{2\pi}{1})$

>0　$\underline{2 \cos(2x)+3}$

range

$[1, 5]$

Domain

$[0, \infty)$

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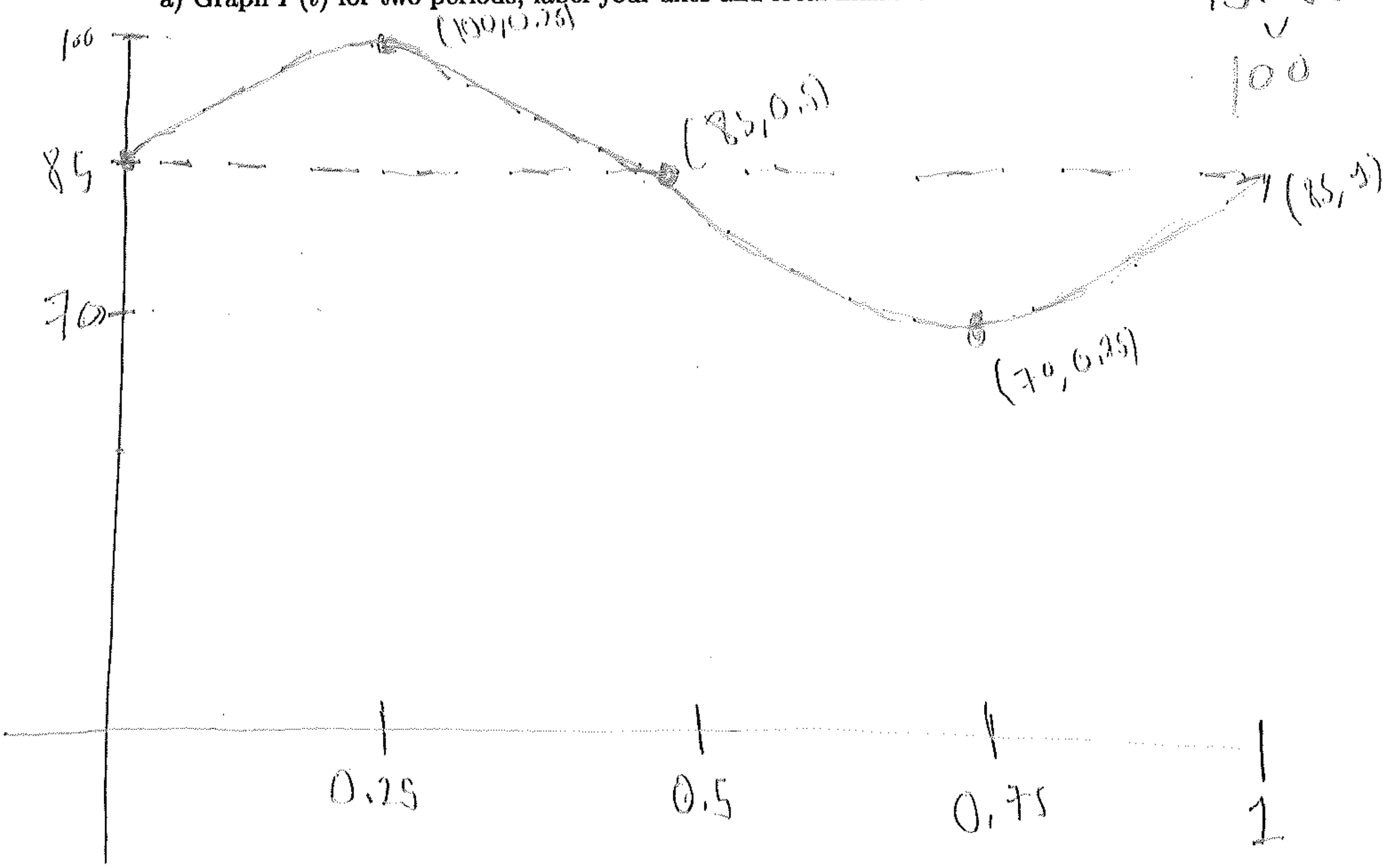
6) Blood pressure is modeled by the function $P(t) = 15\sin(2\pi t) + 85$ where the maximum value of P is the systolic pressure (which is the pressure when the heart beats), the minimum value is the diastolic pressure, and t is time in seconds.

a) Graph $P(t)$ for two periods, label your axes and local maximums.

$15 + 85$
$\downarrow$
100



b) The systolic pressure is <u>100 BPM</u>

c) The diastolic pressure is <u>70 BPM</u>

Midline $= \frac{100+70}{2} = 85$

d) The heart rate is <u>60</u> beats per minute

systolic $= 15 + 85$
$\downarrow$
100

diastolic $= 85 - 15$
$\downarrow$
70

period $= \frac{2\pi}{b} = \frac{2\pi}{2\pi} = 1$

heart rate $= \frac{60}{\text{period}} = \frac{60}{1} = 60$